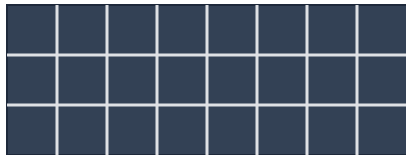


1-6 · Energy budget: dense seed → sparse, task-relevant structure

connectivity mask (rows = hidden units, columns = inputs); a filled cell = a connection

initial: dense seed



energy = 24 connections

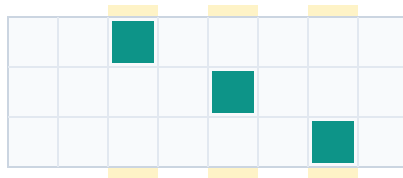
minimize energy



s.t. $\text{acc} \geq \text{target}$

(annealing: prune, rarely grow)

found (SPARSE task)



energy = 3 · 90% on relevant inputs

(chance 33%, drift 34%)

Result: feature selection emerges cleanly. Sparsity and weight-sharing also emerge; the compact aligned filter needs mutation on the shared feature (Sec. 2.3)